

Proofs and Properties of Equality and Congruence

Properties of Equality

Let a , b , and c be real numbers. Then,

Addition Property of Equality:

If $a = b$, then $a + c = b + c$.

Subtraction Property of Equality:

If $a = b$, then $a - c = b - c$.

Multiplication Property of Equality:

If $a = b$, then $ac = bc$.

Division Property of Equality:

If $a = b$, then $\frac{a}{c} = \frac{b}{c}$ if $c \neq 0$.

Reflexive Property of Equality:

$$a = a$$

Symmetric Property of Equality:

If $a = b$, then $b = a$.

Transitive Property of Equality:

If $a = b$, and $b = c$, then $a = c$.

Substitution Property of Equality:

If $a = b$, then b can be substituted for a in any expression.

Properties of Congruence

Examples

Reflexive Property of Congruence:

Figure X is congruent to itself. $\overline{AB} \cong \overline{AB}$, $\angle C \cong \angle C$

Symmetric Property of Congruence:

If figure X is congruent to figure Y , then figure Y is congruent to figure X .
If $\overline{AB} \cong \overline{CD}$, then $\overline{CD} \cong \overline{AB}$. If $\angle E \cong \angle F$, then $\angle F \cong \angle E$.

Transitive Property of Congruence:

If figure X is congruent to figure Y , and figure Y is congruent to figure Z , then figure X is congruent to figure Z .
If $\overline{AB} \cong \overline{CD}$, and $\overline{CD} \cong \overline{EF}$, then $\overline{AB} \cong \overline{EF}$. If $\angle G \cong \angle H$, and $\angle H \cong \angle I$, then $\angle G \cong \angle I$.

Practice

Write the justification for each step below.

① $5x + 1 = 16$	<u>Given.</u>
$5x = 15$	<u>Subt. Prop of =.</u>
$x = 3$	<u>Div Prop of =.</u>

$$\begin{aligned} \textcircled{2} \quad -5 &= \frac{x}{3} - 2 && \underline{\text{Given.}} \\ -3 &= \frac{x}{3} && \underline{\text{Add Prop of =.}} \\ -9 &= x && \underline{\text{Mult. Prop of =.}} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad 2x - 4 &= 4x - 2 && \underline{\text{Given.}} \\ -4 &= 2x - 2 && \underline{\text{Subt. Prop of =.}} \\ -2 &= 2x && \underline{\text{Add Prop of =.}} \\ -1 &= x && \underline{\text{Div Prop of =.}} \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad 9 - 3x &= -6x + 10 && \underline{\text{Given.}} \\ 9 + 3x &= 10 && \underline{\text{Add Prop of =.}} \\ 3x &= 1 && \underline{\text{Subt. Prop of =.}} \\ x &= \frac{1}{3} && \underline{\text{Div Prop of =.}} \end{aligned}$$

Distributive Property of Equality: $a(b+c) = ab+ac$

$$\begin{aligned} \textcircled{5} \quad 2(x-1) + 7 &= 2 && \underline{\text{Given}} \\ 2x - 2 + 7 &= 2 && \underline{\text{Dist Prop of =.}} \\ 2x + 5 &= 2 && \underline{\text{Simplification.}} \\ 2x &= -3 && \underline{\text{Subt. Prop of =.}} \\ x &= -\frac{3}{2} && \underline{\text{Div Prop of =.}} \end{aligned}$$

$$\begin{aligned} \textcircled{6} \quad 5 &= 4 - 3(4x+5) && \underline{\text{Given}} \\ 5 &= 4 - 12x - 15 && \underline{\text{Dist Prop of =.}} \\ 5 &= -11 - 12x && \underline{\text{Simplification.}} \\ 16 &= -12x && \underline{\text{Add Prop of =.}} \\ -\frac{4}{3} &= x && \underline{\text{Div Prop of =.}} \end{aligned}$$

Examples of the Transitive Property of Equality.

I) If $Y=X$ and $Y=L$, then $X=L$.

II) If $\Theta=4$ and $4=\omega$, then $\Theta=\omega$.

III) If $2x-1=7$ and $5x+10=7$, then $2x-1=5x+10$.

IV) If $AB=3$ and $CD=3$, then $AB=CD$.

V) If $m\angle ABD=22^\circ$ and $m\angle DBC=22^\circ$, then $m\angle ABD=m\angle DBC$.

Examples of the Substitution Property of Equality.

I) If $a=6b-4$ and $b=3$, then $a=6(3)-4$.

II) If $AB=x+6$ and $x=1$, then $AB=1+6$.

III) If $m\angle ABC = x^2 + 4x - 3$ and $x=2$, then $m\angle ABC = (2)^2 + 4(2) - 3$.

Proofs

Three Types of Proofs

I) Two-Column Proof

II) Paragraph Proof

III) Flowchart Proof

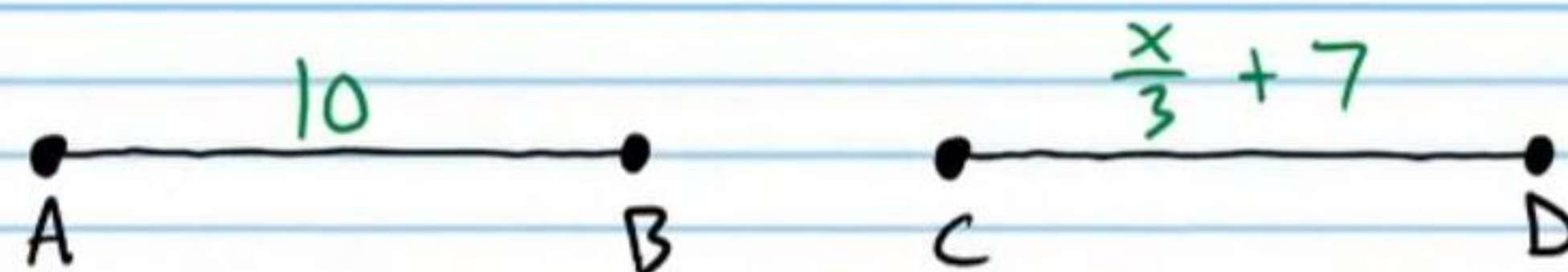
Write a two-column proof for each statement below.

⑦ If the two segments below are congruent, then $x=4$.



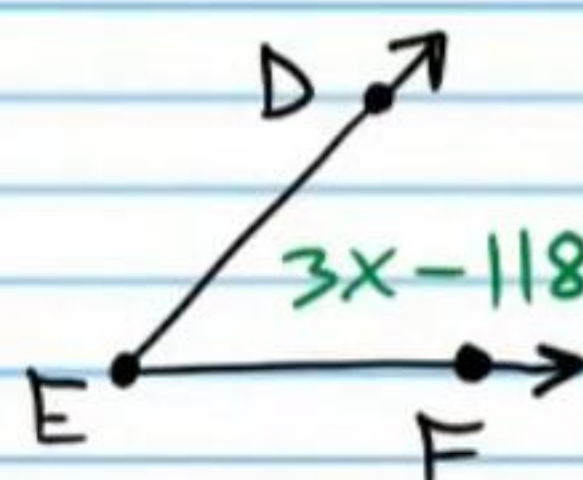
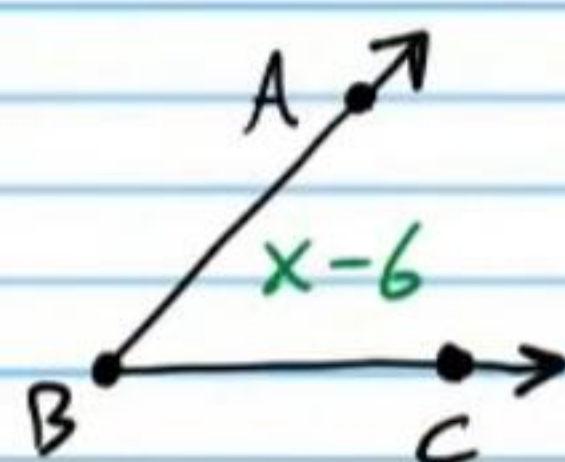
Statement	Reason
I) $\overline{AB} \cong \overline{CD}$	Given.
II) $AB = CD$	Def. of $\cong$ seg
III) $AB = 2x+3$ & $CD = 5x-9$	Given (diagram).
IV) $2x+3 = 5x-9$	Subs. Prop of =
V) $3 = 3x-9$	Subt. Prop of =
VI) $12 = 3x$	Add Prop of =
VII) $4 = x$	Div Prop of =
VIII) $x = 4$	Symm. Prop of =

⑧ If the two segments below are congruent, then $x=9$.



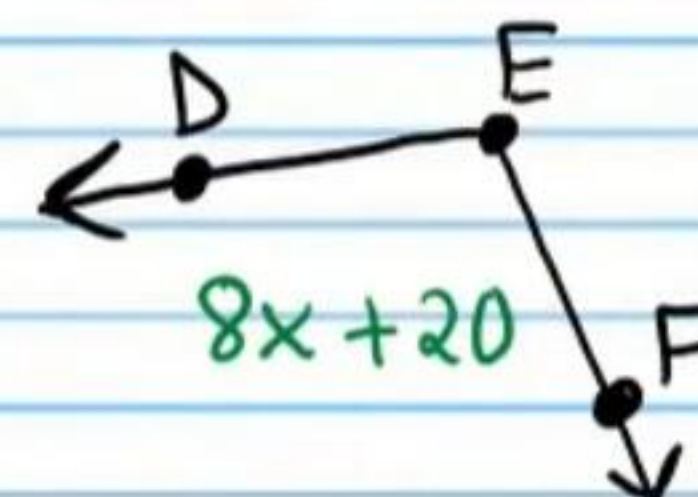
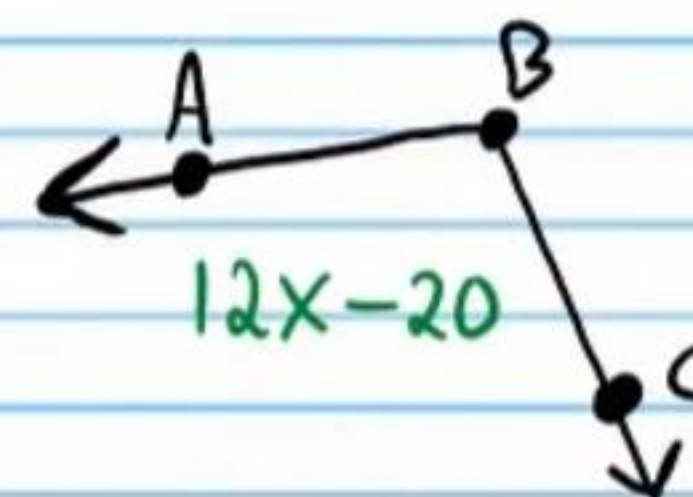
Statement	Reason
I) $\overline{AB} \cong \overline{CD}$	Given.
II) $AB = CD$	Def. of $\cong$ seg.
III) $AB = 10$ & $CD = \frac{x}{3} + 7$	Given (diagram).
IV) $10 = \frac{x}{3} + 7$	Subs. Prop of =
V) $3 = \frac{x}{3}$	Subt. Prop of =
VI) $9 = x$	Mult. Prop of =
VII) $x = 9$	Symm. Prop of =

⑨ If the two angles below are congruent, then $x = 56$.



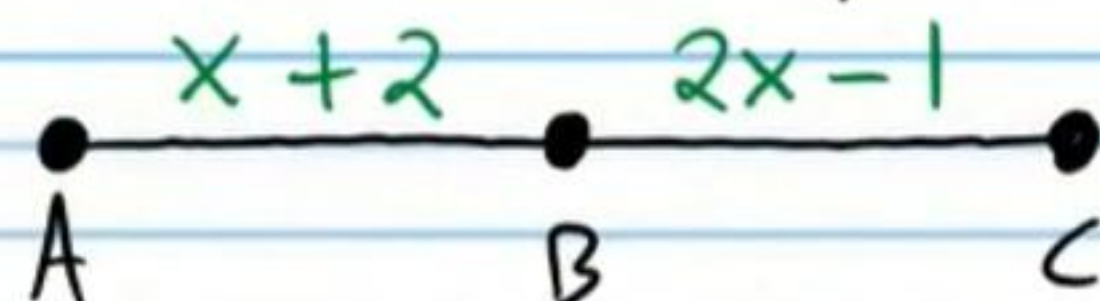
Statement	Reason
I) $\angle ABC \cong \angle DEF$	Given.
II) $m\angle ABC = m\angle DEF$	Def of $\cong$ $\angle$'s.
III) $m\angle ABC = x-6$ and $m\angle DEF = 3x-118$	Given (diagram).
IV) $x-6 = 3x-118$	Subs. Prop of $=$.
V) $-6 = 2x-118$	Subt. Prop of $=$.
VI) $112 = 2x$	Add Prop of $=$.
VII) $56 = x$	Div Prop of $=$.
VIII) $x = 56$	Symm. Prop of $=$.

⑩ If the two angles below are congruent, then $x = 10$.



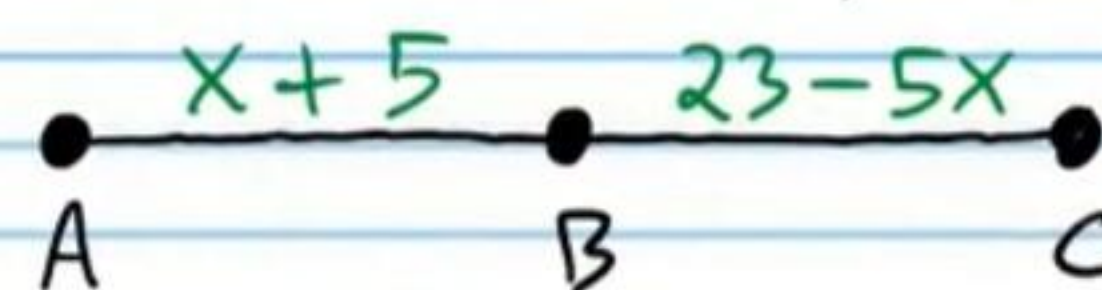
Statement	Reason
I) $\angle ABC \cong \angle DEF$	Given.
II) $m\angle ABC = m\angle DEF$	Def. of $\cong$ $\angle$'s.
III) $m\angle ABC = 12x-20$ & $m\angle DEF = 8x+20$	Given (diagram).
IV) $12x-20 = 8x+20$	Subs. Prop of $=$.
V) $4x-20 = 20$	Subt. Prop of $=$.
VI) $4x = 40$	Add Prop of $=$.
VII) $x = 10$	Div Prop of $=$.

⑪ If point B bisects $\overline{AC}$, then $x = 3$.



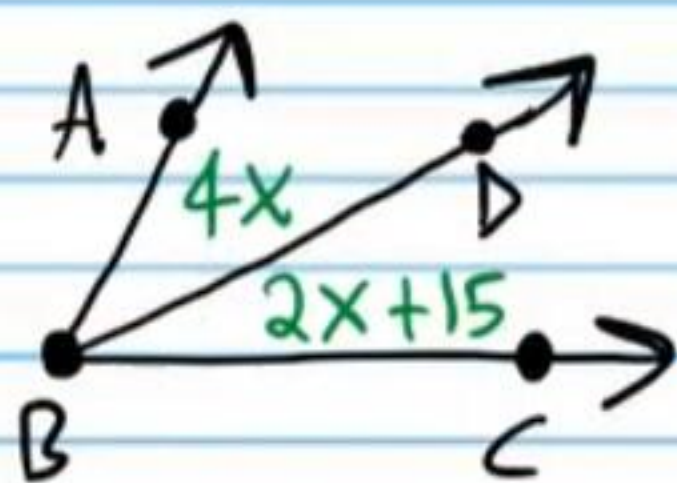
Statement	Reason
I) B bisects $\overline{AC}$	Given.
II) $AB = BC$	Def. of seg bisector
III) $AB = x + 2$ & $BC = 2x - 1$	Given (diagram).
IV) $x + 2 = 2x - 1$	Subs. Prop of =.
V) $2 = x - 1$	Subt. Prop of =.
VI) $3 = x$	Add Prop of =.
VII) $x = 3$	Symm. Prop of =.

⑫ If point B bisects $\overline{AC}$, then $x = 3$.



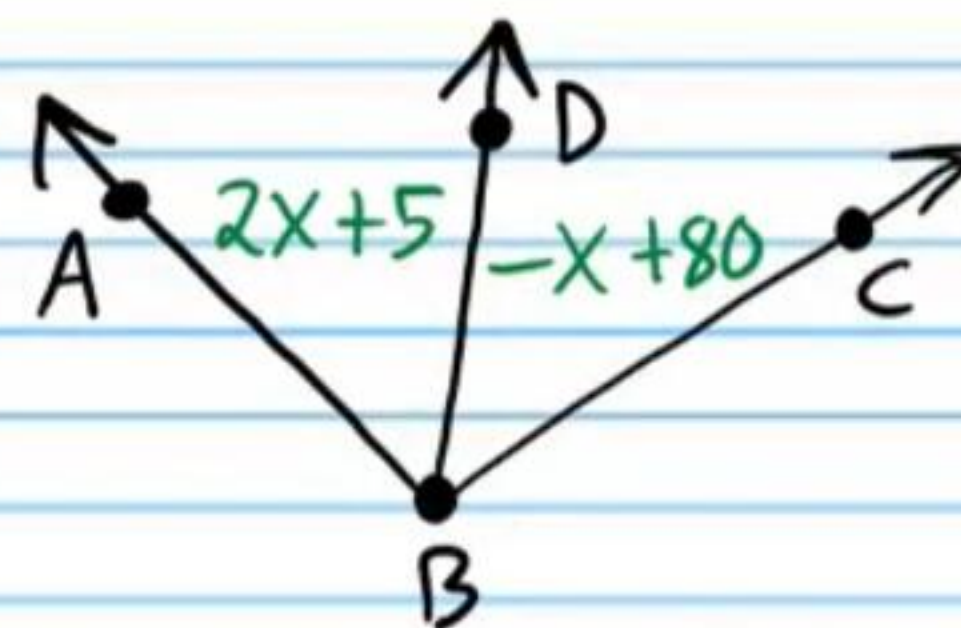
Statement	Reason
I) B bisects $\overline{AC}$.	Given.
II) $AB = BC$	Def. of seg bisector.
III) $AB = x + 5$ & $BC = 23 - 5x$	Given (diagram).
IV) $x + 5 = 23 - 5x$	Subs. Prop of =.
V) $6x + 5 = 23$	Add Prop of =.
VI) $6x = 18$	Subt. Prop of =.
VII) $x = 3$	Div. Prop of =.

⑬ If $\overrightarrow{BD}$ bisects $\angle ABC$, then $x = 7.5$.



Statement	Reason
I) $\overrightarrow{BD}$ bisects $\angle ABC$.	Given.
II) $m\angle ABD = m\angle DBC$	Def. of $\angle$ bisector
III) $m\angle ABD = 4x$ and $m\angle DBC = 2x + 15$	Given (diagram).
IV) $4x = 2x + 15$	Subst. Prop of =.
V) $2x = 15$	Subst. Prop of =.
VI) $x = 7.5$	Div Prop of =.

⑭ If $\overrightarrow{BD}$ bisects $\angle ABC$, then $x = 25$.



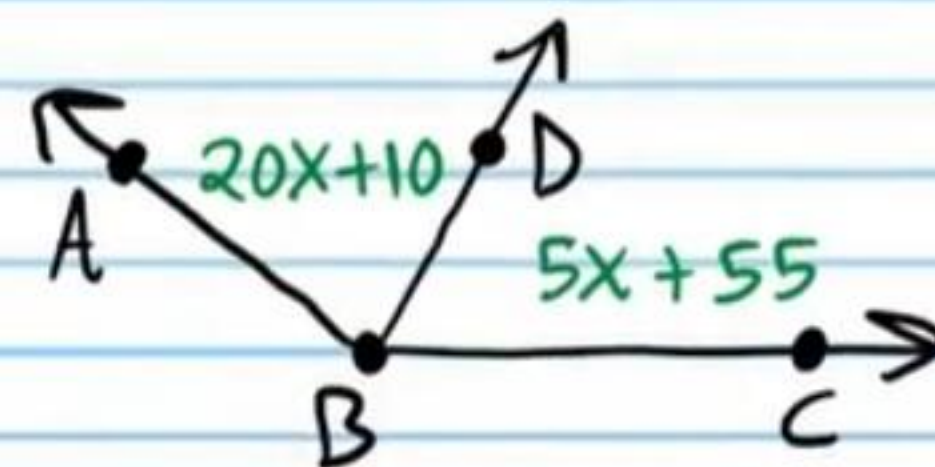
Statement	Reason
I) $\overrightarrow{BD}$ bisects $\angle ABC$.	Given.
II) $m\angle ABD = m\angle DBC$	Def. of $\angle$ bisector.
III) $m\angle ABD = 2x + 5$ and $m\angle DBC = -x + 80$.	Given (diagram)
IV) $2x + 5 = -x + 80$	Subst. Prop of =.
V) $3x + 5 = 80$	Add Prop of =.
VI) $3x = 75$	Subst. Prop of =.
VII) $x = 25$	Div Prop of =.

⑮ If $\overline{AB} \cong \overline{CD}$ below, then $AB = 12$.



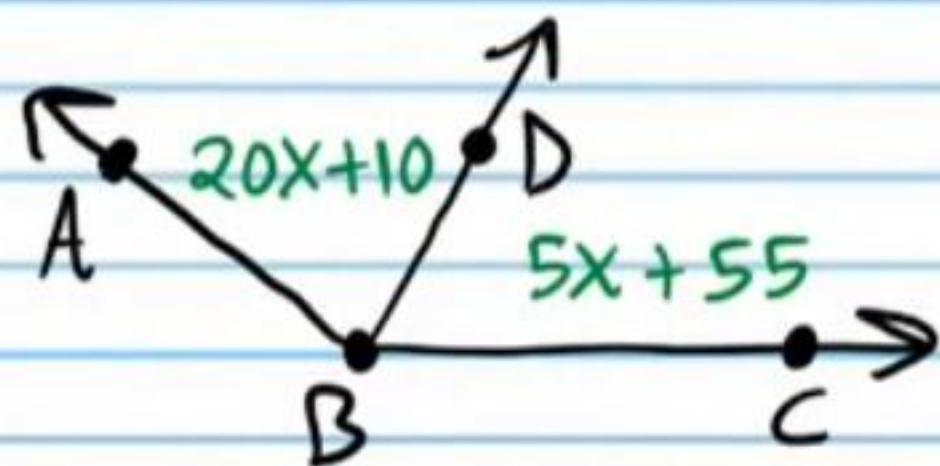
Statement	Reason
I) $\overline{AB} \cong \overline{CD}$	Given.
II) $AB = CD$	Def. of $\cong$ seg.
III) $AB = 3x - 6$ & $CD = 2x$	Given.
IV) $3x - 6 = 2x$	Subst. Prop of =.
V) $x - 6 = 0$	Subst. Prop of =.
VI) $x = 6$	Add Prop of =.
VII) $AB = 3(6) - 6$	Subst. Prop of =.
VIII) $AB = 12$	Simplification.

⑯ If $\overrightarrow{BD}$ bisects $\angle ABC$, then $m\angle DBC = 70^\circ$.



Statement	Reason
I) $\overrightarrow{BD}$ bisects $\angle ABC$	Given.
II) $m\angle ABD = m\angle DBC$	Def. of $\angle$ bisector.
III) $m\angle ABD = 20x + 10$ and $m\angle DBC = 5x + 55$.	Given (diagram).
IV) $20x + 10 = 5x + 55$	Subst. Prop of =.
V) $15x + 10 = 55$	Subst. Prop of =.
VI) $15x = 45$	Subst. Prop of =.
VII) $x = 3$	Div. Prop of =.
VIII) $m\angle DBC = 5(3) + 55$	Subst. Prop of =.
IX) $m\angle DBC = 70^\circ$	Simplification.

(17) The converse of #16.



Statement	Reason
I) $m\angle DBC = 70^\circ$	Given.
II) $m\angle DBC = 5x + 55$	Given (diagram)
III) $70 = 5x + 55$	Trans. Prop of =.
IV) $15 = 5x$	Subst. Prop of =.
V) $3 = x$	Div Prop of =.
VI) $m\angle ABD = 20x + 10$	Given (diagram).
VII) $m\angle ABD = 20(3) + 10$	Subst. Prop of =.
VIII) $m\angle ABD = 70^\circ$	Simplification.
IX) $m\angle DBC = m\angle ABD$	Trans. Prop of =.
X) $\overrightarrow{BD}$ bisects $\angle ABC$.	Def. of $\angle$ bisector.

(18) The converse of #7.



Statement	Reason
I) $x = 4$	Given.
II) $AB = 2x + 3$ & $CD = 5x - 9$	Given (diagram).
III) $AB = 2(4) + 3$ & $CD = 5(4) - 9$	Subst. Prop of =.
IV) $AB = 11$ & $CD = 11$	Simplification.
V) $AB = CD$	Trans. Prop of =.
VI) $\overline{AB} \cong \overline{CD}$	Def of $\cong$ seg